SUYOG CHOUGULE

Bangalore, Karnataka, India • suyogchougule800@gmail.com • +91 9741687782
LinkedIn • GitHub

Education

Manipal Institute of Technology Manipal, India
B.Tech in Mechatronics | 2020 – 2024 | **CGPA: 8.3/10**

SV PU College Bangalore, India
High School | Grad. May 2020| **Grade: 91.3/100**

Experience

Twara Robotics (formerly Gati Robotics) Bangalore, India
Associate Engineer - Embedded Systems July 2024 – Present

- Engineered a custom backend for robotic arms, optimizing real-time code to reduce CAN latency from 500ms(from bash script) to under 200ms through improved cpp algorithms .
- Developing high-performance embedded systems for robotic controllers, optimizing CAN-based communication for real-time control and reliability.

Embedded Systems Intern (Offline Mode) Jan 2024 – June 2024

- Worked on ROS2 control interfaces for robotic systems.
- Incorporated MoveIt2 for motion planning and trajectory optimization.

Embedded Systems Intern (Online Mode) Aug 2023 – Dec 2023

- Developed generic ROS2 driver for Twara Actuators.
- Control software for Soft robotics testing rig, by controlling Dynamixel motors.

Projects

Base Board for Twara Robotic Arms

- **State Machine Control:** Designed and developed an STM32-based board to manage the robotic arm's startup sequence—powering the 24V SMPS, restoring the last motor turn count (LMTC), initializing motor drivers, and controlling all onboard I/Os.
- **Motor Position Retention and Communication:** Implemented EEPROM-based LMTC storage, enabling motor position recovery after power resets due to single-turn absolute encoders. Integrated CAN and RS485 communication for seamless data exchange with motor drivers.
- **Power Efficiency and I/O Handling:** Engineered a low-power architecture with sleep mode when the robotic arm was off on battery. Designed the board with 8× 24V DOs, 8× DIs (0-10V), and 8× AIs (0-9V/24V) to support various sensors and actuators.

GUI for Twara Robotic Arm

- Developed a **digital twin simulation** and virtual arm visualization using **Three.js**.
- Integrated Three.js with Node-RED for real-time motion planning feedback.

Twara USB to CAN Converter

- Developed a **USB to CAN converter** supporting **SocketCAN on Linux** and other OS tools.

ROS2 Control Driver for Twara Actuators

- Developed a **ROS2 Control hardware interfaces/Drivers** for Twara motor drivers, AMR kit, and **robotic arms.**
- Integrated **MoveIt2 for real-time servoing** and trajectory planning.

Personal Projects

Ripple Technologies (Startup) Manipal, India
Founder Nov 2021 – Dec 2024

- Secured **7 lakh funding** under the Nidhi Prayas grant from the Department of Science, Govt. of India, for Type 3 DC charger development. Raised additional **1 lakh rupees** for initial prototype from collage.
- Designed **50kW Type 3 DC Charger**, integrating Vienna Rectifier and **PSFB DC-DC Converter**.
- Developed PSFB Converter and Vienna Rectifier **driver circuits**(PCB Design) **and firmware** with STM32-based MCU, voltage/current sensing, and IGBT/SiC drivers.
- Implemented real-time control and fault detection in PSFB converter firmware.

Skills

Programming: C, C++, Python, JS
Embedded Systems: Baremetal Programming, STM32 HAL programming, FreeRTOS, Arduino, Raspberry Pi
PCB Design: High-Power PCB design, multi-layer PCB design (KiCad, Fusion 360 Electronic Design)
Control Systems: Motor control (PWM, PID), PID tuning, MoveIt2, ROS2 Control, Fault Detection and Energy Optimization, Field Oriented Control
Communication Protocols: CAN, RS485, I2C, SPI
Simulation & Modeling: MATLAB(Simscape), Gazebo, Three.js
Power Electronics: Design of driver circuits and developement of embedded firmware, DC-DC Converters, Inverters, SiC Mosfets/IGBT Drivers.

Talks & Recognition

- **Speaker at ROSCON India 2024 (IISc Bangalore):** Delivered a talk on **ROS2 Control in Mobile Robots and Robotic Arms**.
- State topper in 12th board mathematics exam (scored 100/100).